

Use of AI

This document lists the instances where AI tools were used to support the coding in this project. For each use, the required fields (Tool, Prompt, and Output Summary) are provided.

1. Assistance in Implementing the Return Horizon Loop

- a. **Tool:** ChatGPT
- b. **Prompt:** "I want to loop over prediction horizons (1, 3, 7 days) and store the results for each model. Can you show me a clean way to structure this in Python?"
- c. **Output (Summary):** ChatGPT provided an example of iterating through horizons and appending results to a dictionary/DataFrame. I used the structural idea but wrote my own version to match the notebook's variables and workflow.

2. Help Correcting Alignment Issues Between Features and Targets

- a. **Tool:** ChatGPT
- b. **Prompt:** "My shifted return columns don't line up, and some models are training on NaNs. How should I handle this?"
- c. **Output (Summary):** ChatGPT suggested using `dropna()` after shifting targets and verifying index alignment. I applied the suggestion to fix target/feature alignment in the notebook.

3. Guidance Fixing Errors in Ridge and XGBoost Setup

- a. **Tool:** ChatGPT
- b. **Prompt:** "I'm getting inconsistent shape errors when training Ridge Regression and XGBoost across multiple horizons and filtering levels. Can you help me debug how the feature matrix and target vectors should be structured so the models train consistently?"
- c. **Output (Summary):** ChatGPT walked through how model inputs should be organized when iterating over different horizons, including verifying index alignment after shifting returns, ensuring matching row counts after filtering, and converting DataFrames to arrays only after confirming shape consistency. This helped me correct shape mismatches and stabilize the model-training blocks in both notebooks.

4. Writing a Clean Metric Evaluation Block

- a. **Tool:** ChatGPT
- b. **Prompt:** "I need a simple block of code to compute R^2 and correlation for each model. Can you give an example?"

- c. **Output (Summary):** ChatGPT returned a short example using `r2_score` and `np.corrcoef`. I used a variation of this pattern for the evaluation section of the notebooks.

5. Generating a Starter TCN Architecture (Torch)

- a. **Tool:** ChatGPT
- b. **Prompt:** “Can you write a PyTorch implementation of a TCN block?”
- c. **Output (Summary):** ChatGPT generated a minimal TCN class with residual layers and a forward function. I used this as the initial template and customized it for my dataset and feature shapes.

6. Converting My Final Written Report to LaTeX

- a. **Tool:** ChatGPT
- b. **Prompt:** “Convert my Google Docs report into LaTeX formatting while preserving section structure and spacing.”
- c. **Output (Summary):** ChatGPT created a base LaTeX document including standard packages, section headings, and placeholder figure environments. I used this as a formatting scaffold.